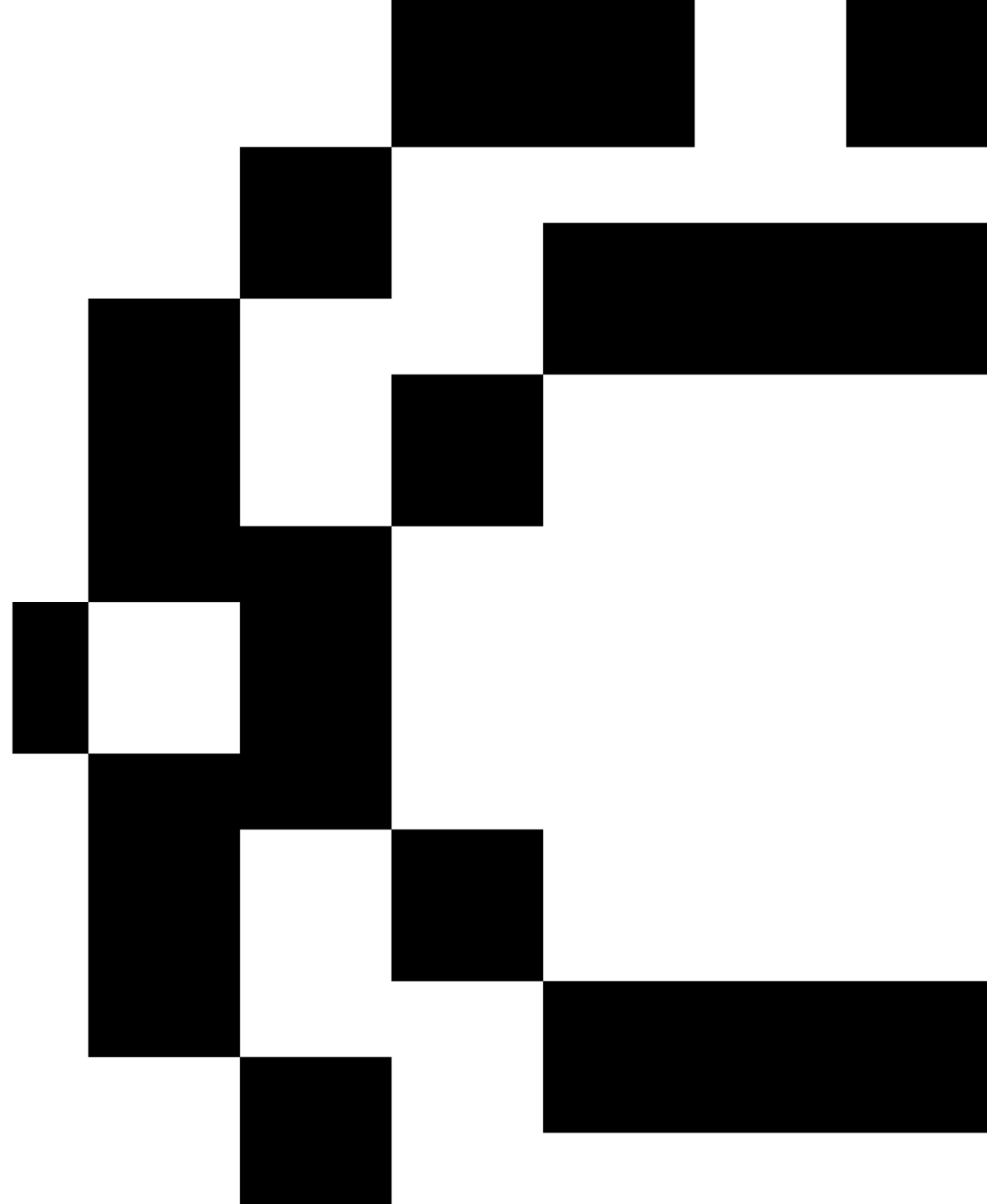


Big Data with PySpark

Three use cases. Three datasets.
One library.

Group 10 Big Data Analytics Project

Bárbara Franco • Catarina Mendinhas • Maria Fonseca •
Rodrigo Santos • Rodrigo Teixeira



The challenge

Three different datasets today, and at scale tomorrow.

Baldness Classification

- Thousands of images
- Severe class imbalance and visual variability

01

Player's Market Value Prediction

- Structured records
- Dozens of features per row

02

Money Laundering Detection

- Millions of transactions
- Hidden patterns in networks

03

01. Baldness Classification

NOT BALD

True: NotBald
Pred: NotBald
Conf: 0.00



True: NotBald
Pred: NotBald
Conf: 0.00



True: Bald
Pred: Bald
Conf: 1.00



BALD

True: Bald
Pred: Bald
Conf: 0.99



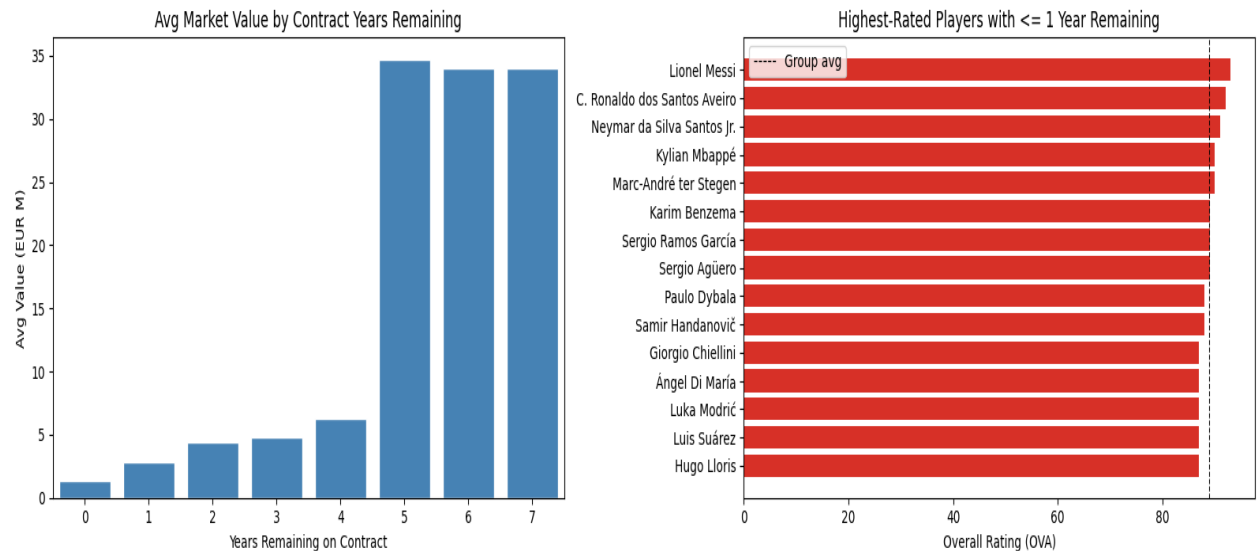
WHAT ABOUT MESSI?

NOT BALD



02. Player's Market Value Prediction

Contract Expiry Risk Analysis



Test set prediction · L. Messi

Name	Age	OVA	Actual value	Predicted value
L. Messi	33	93	€103.5M	€116.5M

Error: +€13.0M (+12.6%)

Built end-to-end in PySpark

Explore

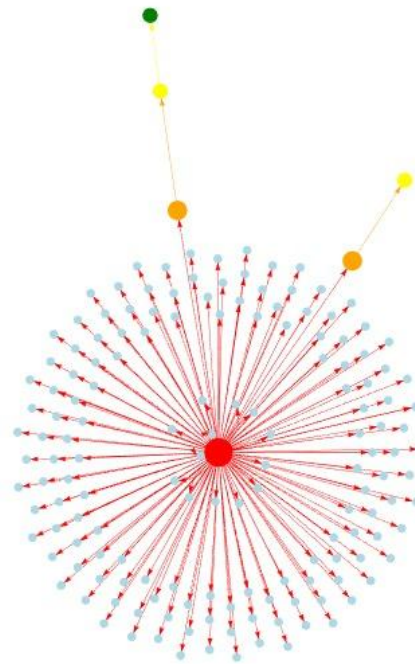
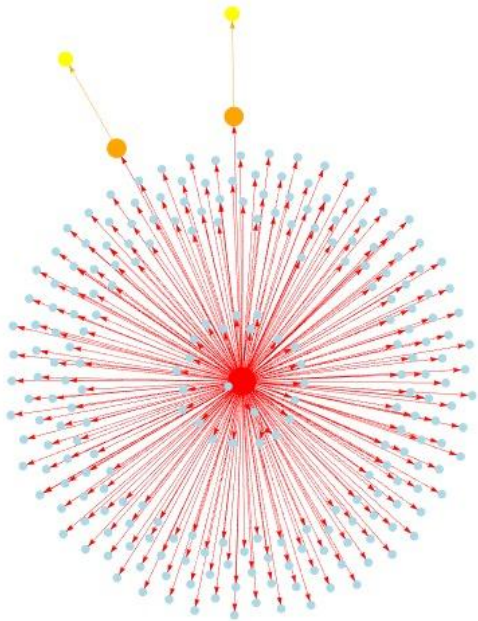


Prepare



Train ML pipeline

03. Money Laundering Detection



- **Smurfing:** 2 distributors disperse laundering transactions across ~400 accounts.
- **Cyclic:** small group of accounts repeatedly circulates funds inside a near-closed loop.
- **Streaming:** flags accounts in real time.

Key results

Three problems, three outcomes — one library.

Baldness classification

97%

Accuracy

Player's Market Value Prediction

207k€

Error

Fraud-Network Analysis

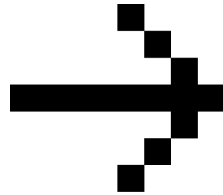
5M

Transactions

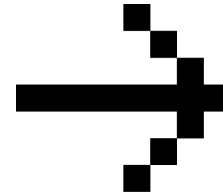
Why Spark?

The four V's of big data — one concrete answer for each.

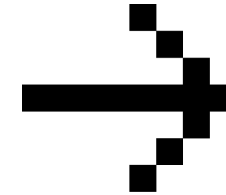
01



02



03



04

Volume

From 5 million records today to 500 million tomorrow.

The same code runs at any scale.

Velocity

Real-time streaming raises alerts as data arrives, not hours later.

Variety

Tables and images, radically different data types, one platform.

Veracity

Reproducible pipelines with clear data quality checks and auditable models.

What the client gains

01



Three working prototypes

Value prediction, image classification and fraud detection, all addressing real business needs.

02



Architecture built to scale

The same PySpark approach runs on small environments today and on much larger infrastructure tomorrow, with no redesign.

03



One unified platform

Data prep, analytics, ML, graph processing and streaming in one stack, fewer tools, simpler path to production.

Faster decisions, ready to scale

Process data at scale, build reliable pipelines, answer business questions faster.

Conclusion.



Three cases. Three datasets. One platform.

PySpark scales when your data does.

Same code: from 19k rows to 5M transactions.

Real-time streaming on top. Alerts as data arrives.

Ready for production: one tech stack to maintain.

Thank you.